Lab 9

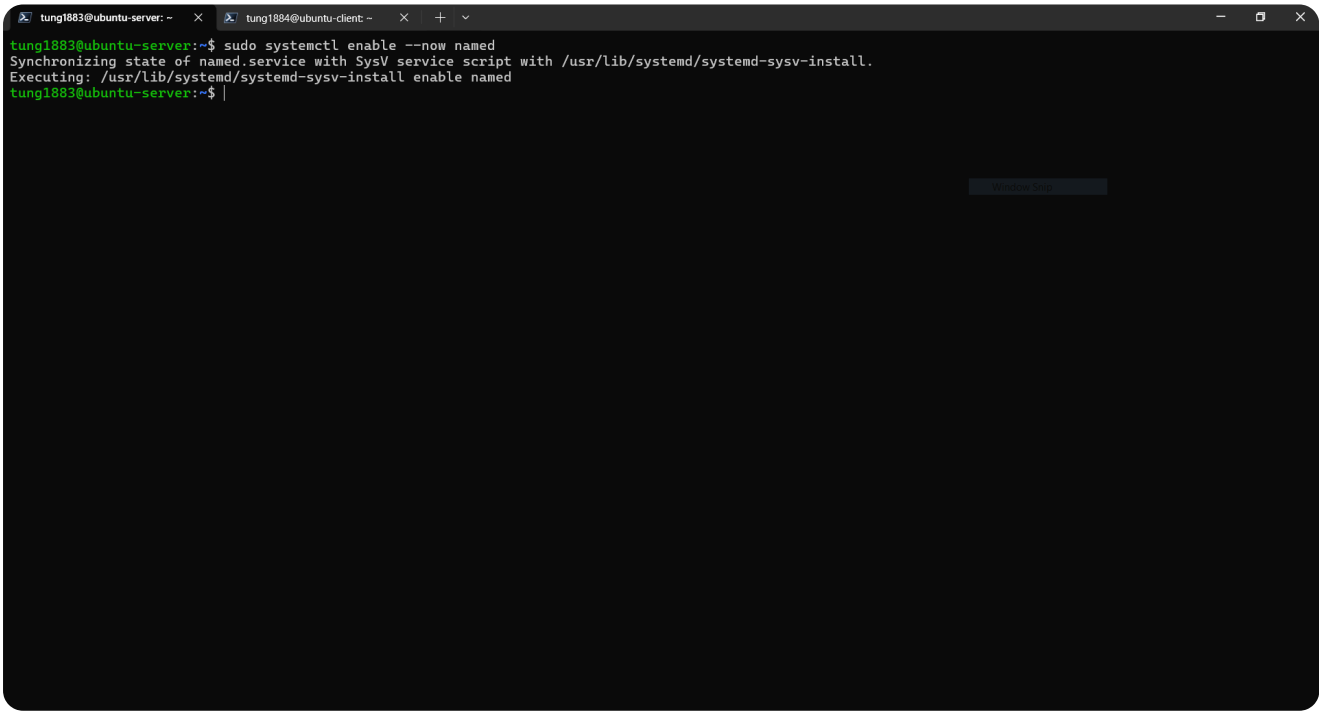
Họ và tên: Nguyễn Đức Tùng – MSV: 21020556

- Máy chủ DNS/DHCP `ubuntu-server`, IP tĩnh `192.168.139.10/24` (em xin phép thầy sử dụng lại IP từ Lab8 thay vì dải `192.168.10.0/24` ạ!!)
- Máy khách `ubuntu-client` với user tung1884, sẽ nhận IP qua DHCP ở mục 4
- Default gateway: `192.168.139.1` – Tên miền thực hành: `lab.local`

1. Cấu hình DNS

1.1 Cài đặt và khởi động dịch vụ DNS

```
sudo apt update
sudo apt install -y bind9 bind9utils bind9-doc dnsutils
sudo systemctl enable --now named
```



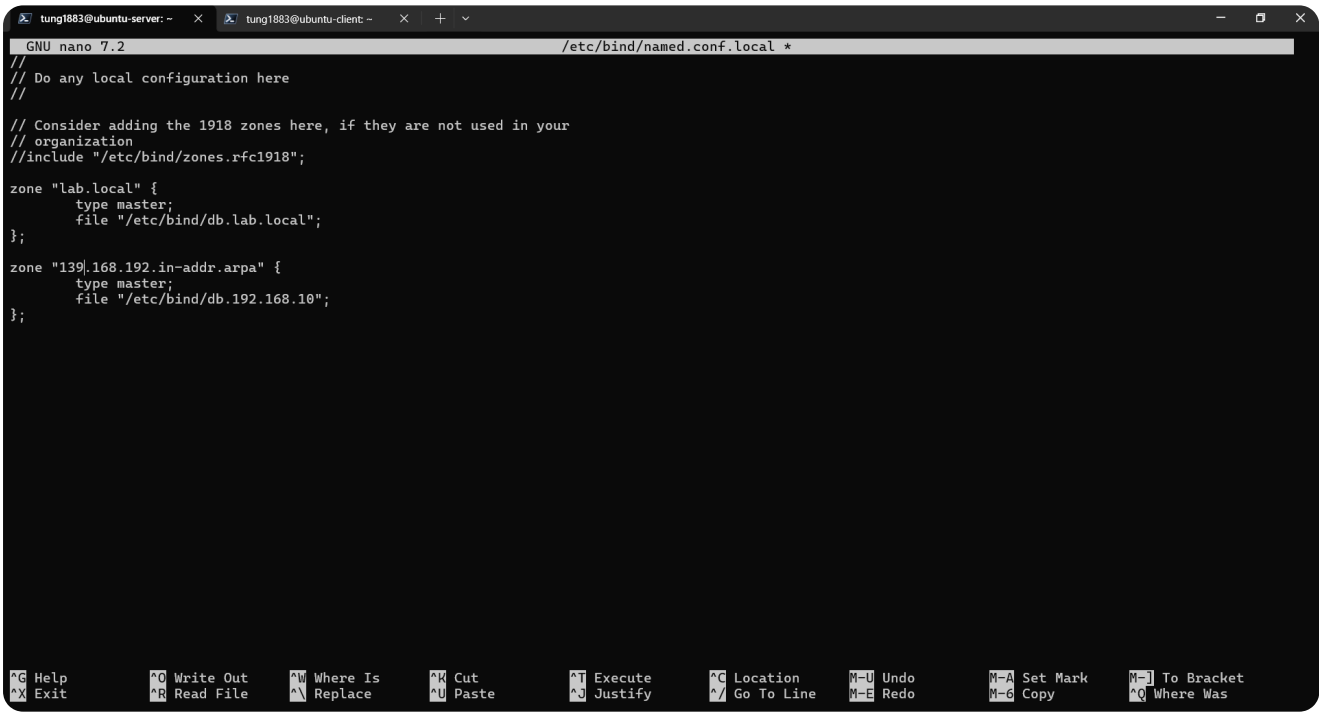
1.2 Khai báo vùng phân giải thuận + ngược

```
sudo nano /etc/bind/named.conf.local
```

Thêm vào cuối file:

```
zone "lab.local" {
    type master;
    file "/etc/bind/db.lab.local";
};

zone "139.168.192.in-addr.arpa" {
    type master;
    file "/etc/bind/db.192.168.139";
};
```

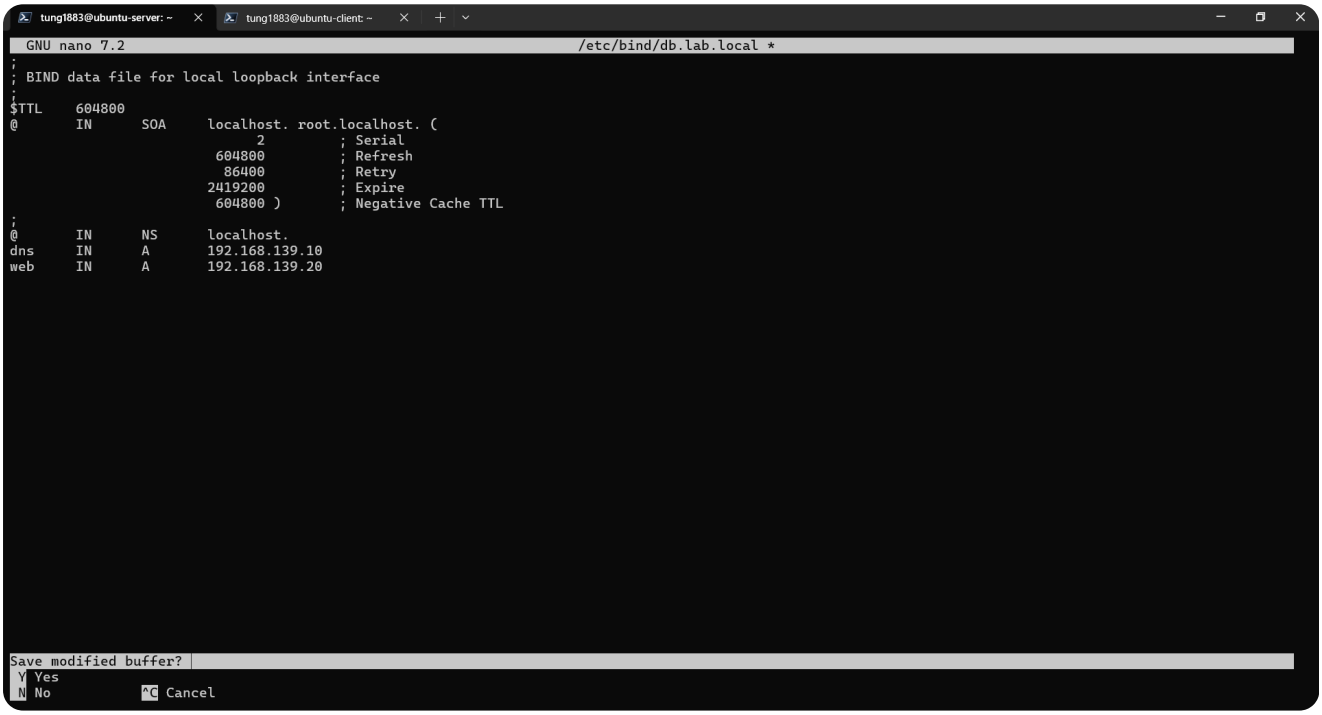


1.3 Tạo file zone thuận (db.lab.local) với 2 bản ghi A

```
sudo cp /etc/bind/db.local /etc/bind/db.lab.local
sudo nano /etc/bind/db.lab.local
```

Edit file thành:

```
$TTL    604800
@       IN      SOA     dns.lab.local. admin.lab.local. (
                              2         ; Serial
                         604800         ; Refresh
                          86400         ; Retry
                        2419200         ; Expire
                         604800 )       ; Negative Cache TTL
;
@       IN      NS      dns.lab.local.
dns     IN      A       192.168.139.10
web     IN      A       192.168.139.20
```



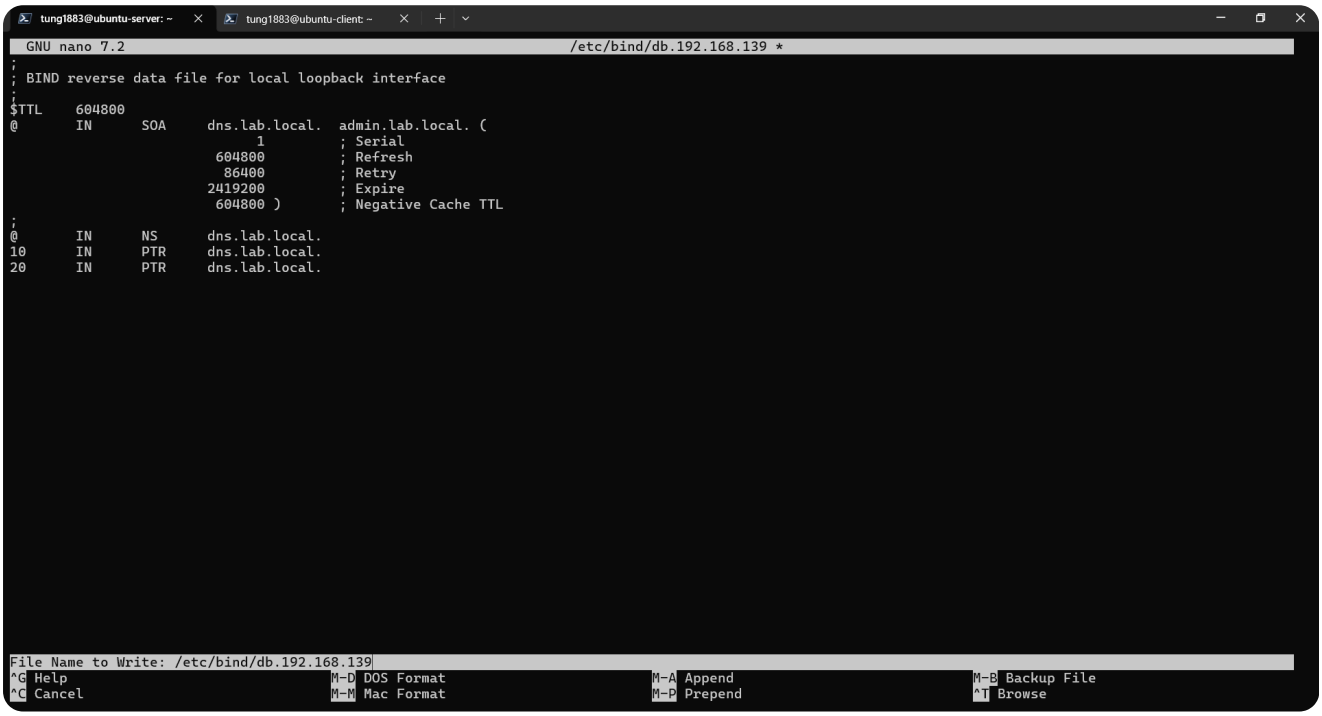
1.4 Tạo file zone ngược (db.192.168.139) với bản ghi PTR

```
sudo cp /etc/bind/db.127 /etc/bind/db.192.168.139
sudo nano /etc/bind/db.192.168.139
```

Edit file thành:

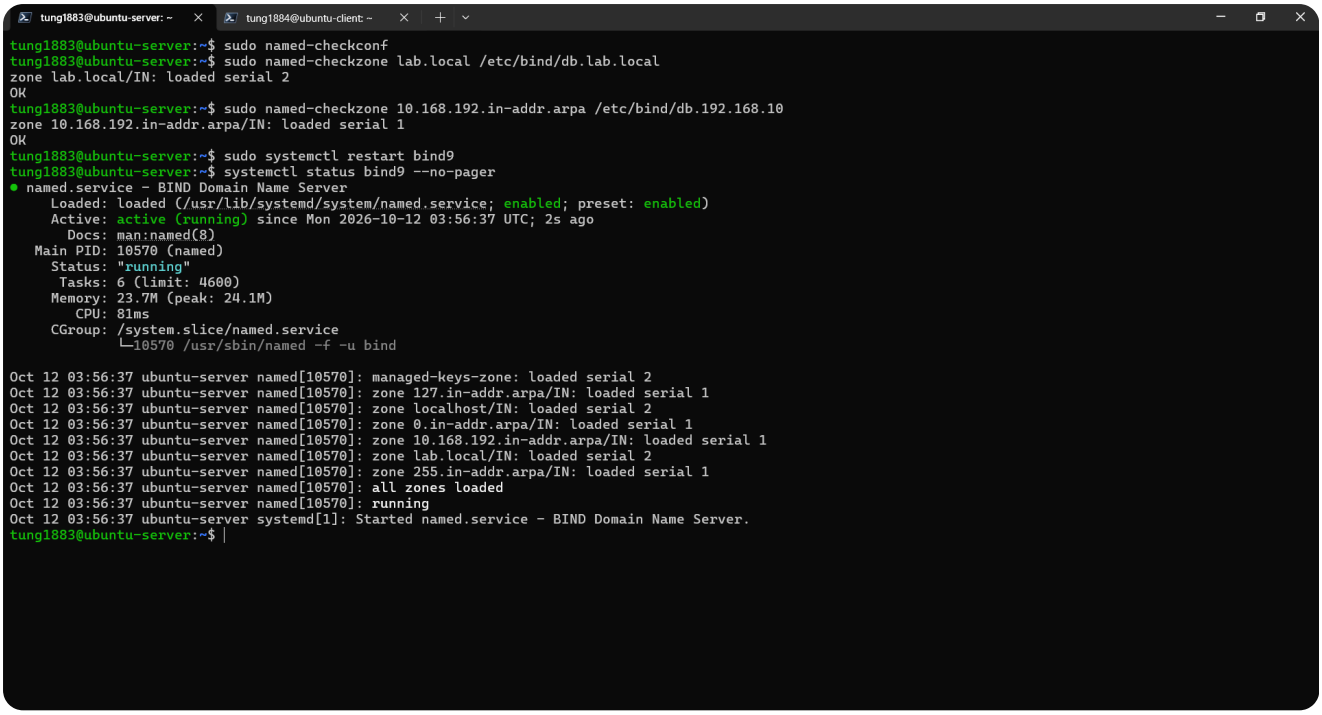
```
$TTL    604800
@       IN      SOA     dns.lab.local. admin.lab.local. (
                              1         ; Serial
```

```
                          604800          ; Refresh
                           86400          ; Retry
                         2419200          ; Expire
                          604800 )        ; Negative Cache TTL
;
@       IN      NS      dns.lab.local.
10      IN      PTR     dns.lab.local.
20      IN      PTR     web.lab.local.
```



1.5 Kiểm tra cú pháp trước khi khởi động lại

```
sudo named-checkconf
sudo named-checkzone lab.local /etc/bind/db.lab.local
sudo named-checkzone 139.168.192.in-addr.arpa /etc/bind/db.192.168.139
sudo systemctl restart named
systemctl status named --no-pager
```

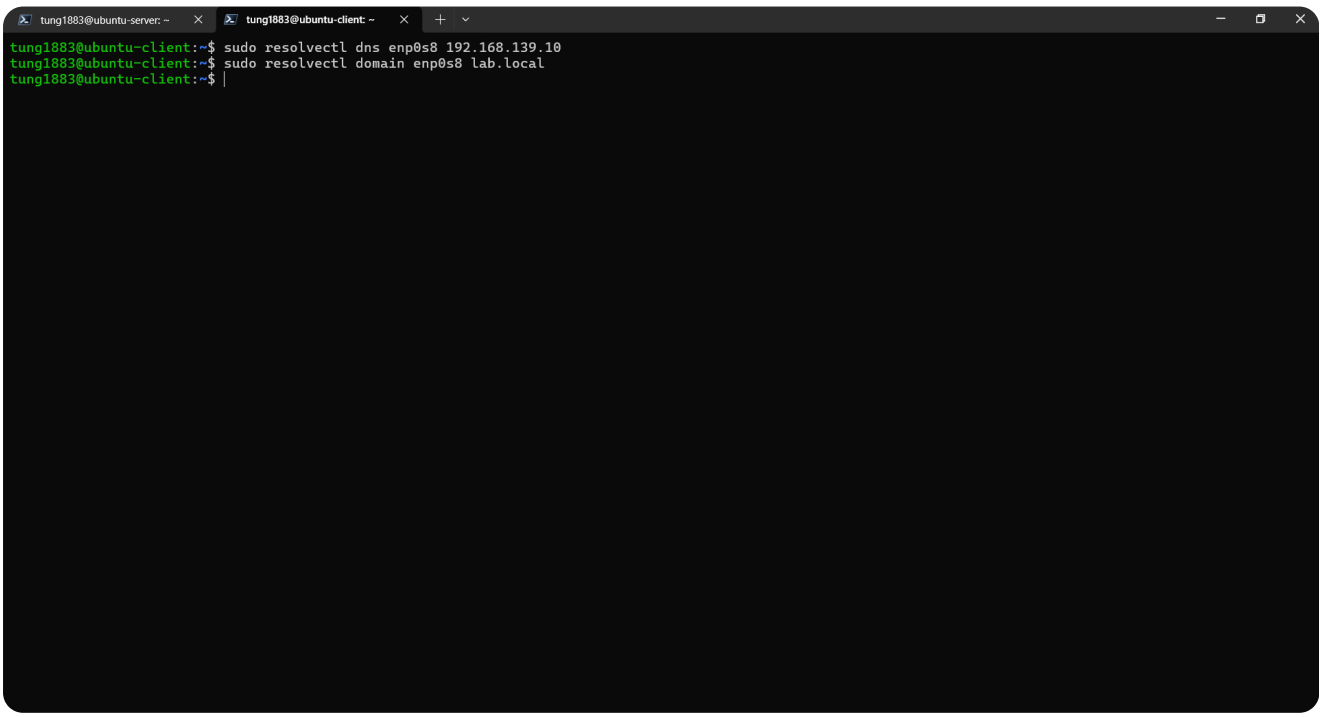


2. Kiểm tra DNS (từ máy khách)

2.1 Trỏ máy khách về DNS server (tạm thời, trước khi cấu hình DHCP)

```
sudo resolvectl dns enp0s8 192.168.139.10

sudo resolvectl domain enp0s8 lab.local
```



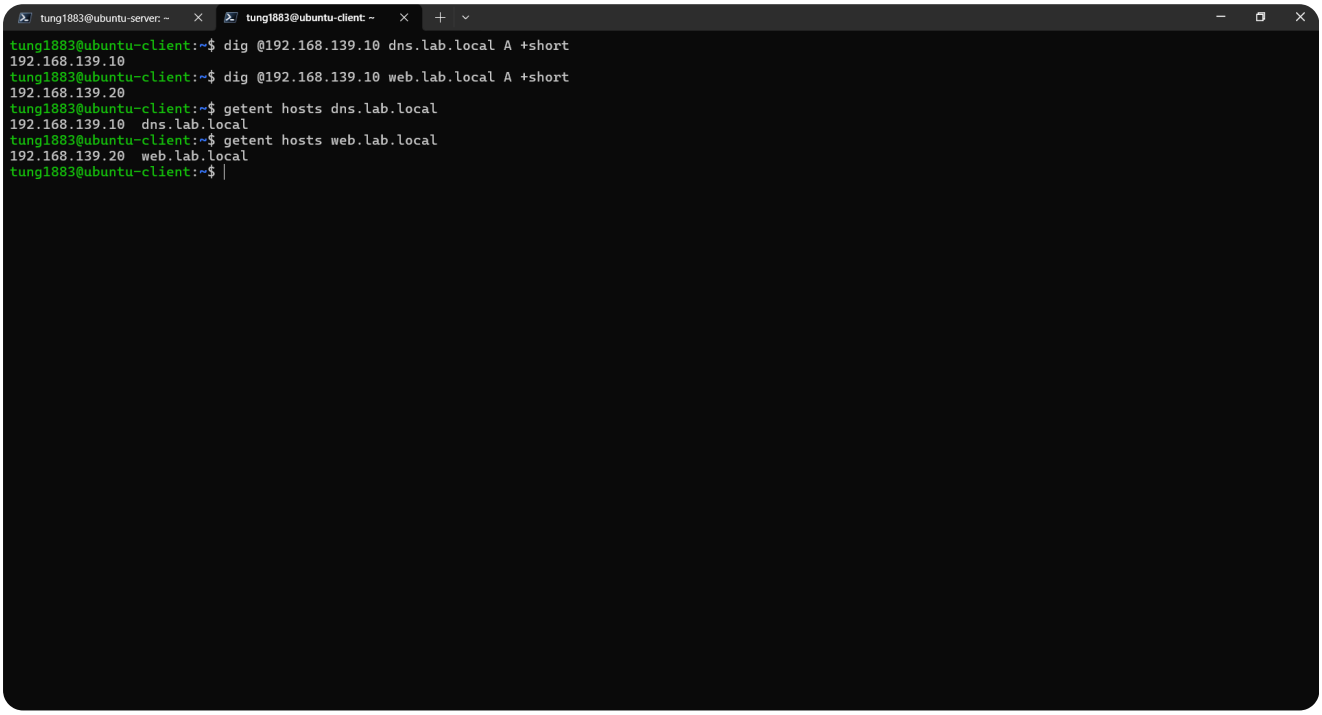
2.2 Phân giải thuận

```
dig @192.168.139.10 dns.lab.local A +short

dig @192.168.139.10 web.lab.local A +short

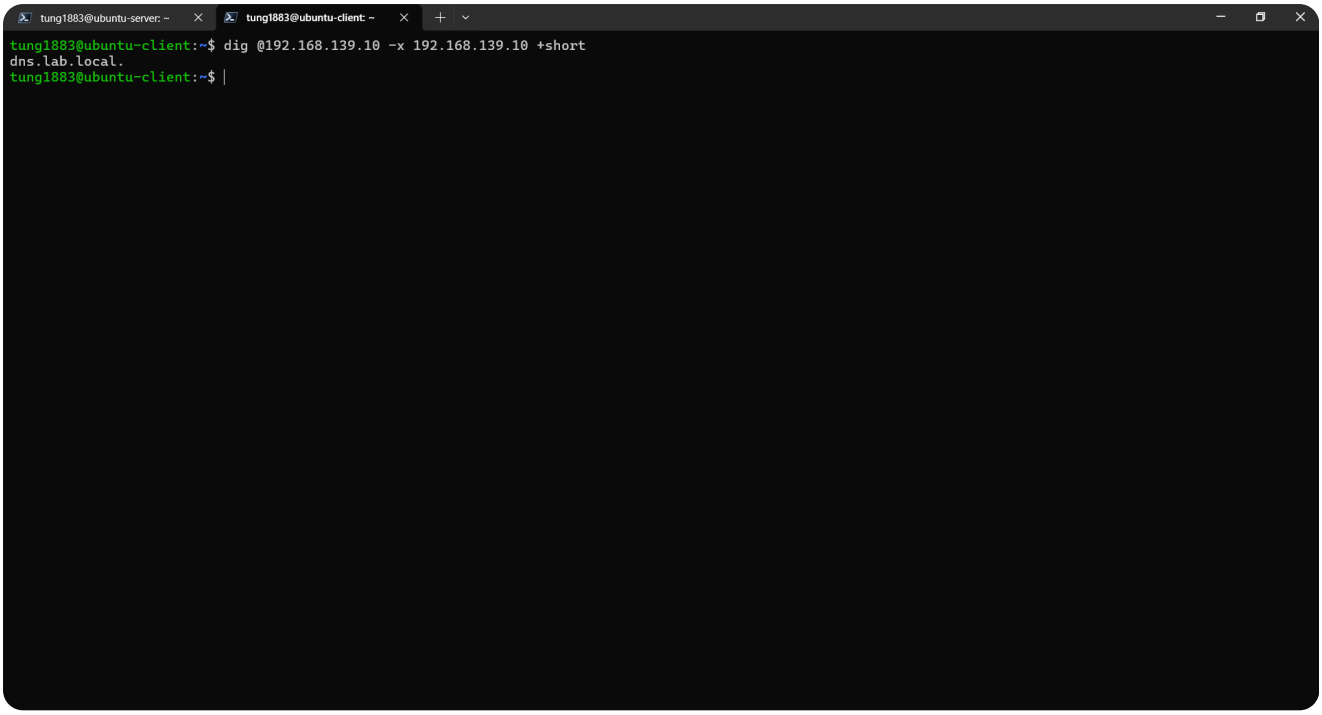
getent hosts dns.lab.local

getent hosts web.lab.local
```



2.3 Phân giải ngược

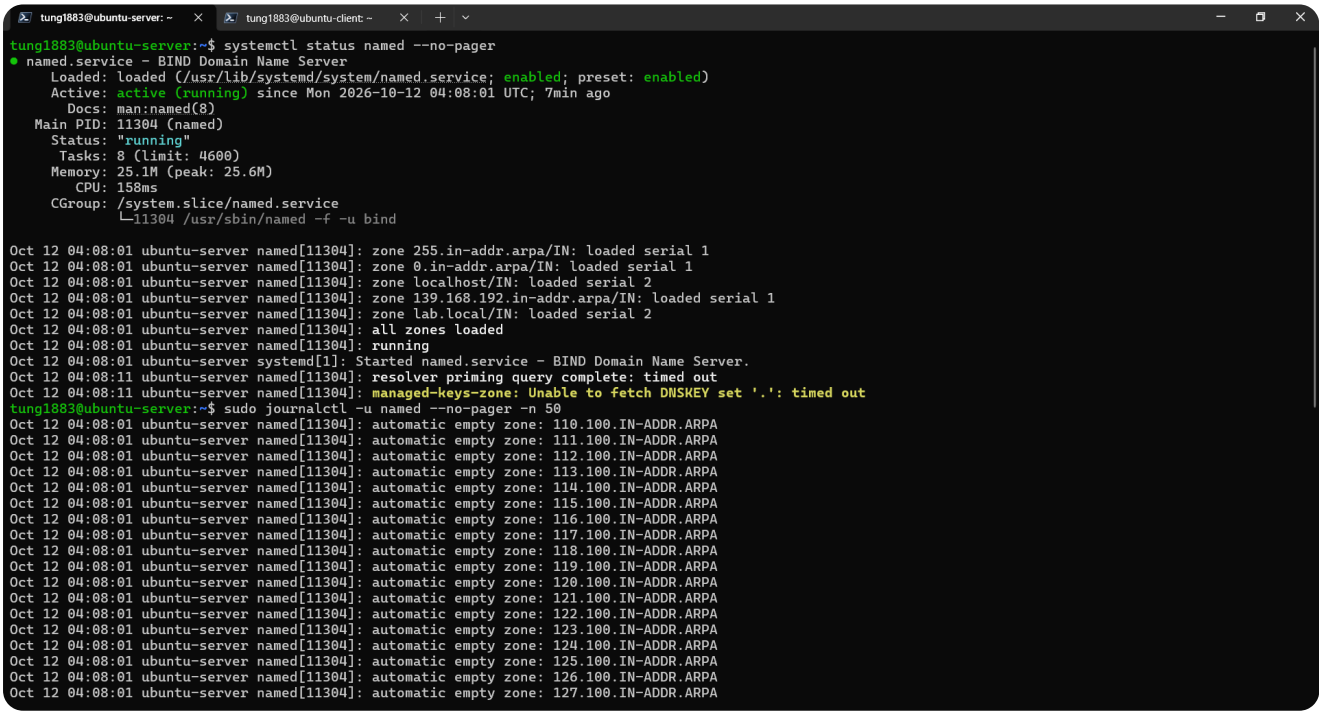
```
dig @192.168.139.10 -x 192.168.139.10 +short
```



2.4 Trạng thái và nhật ký dịch vụ DNS (trên server)

```
systemctl status named --no-pager

sudo journalctl -u named --no-pager -n 50
```

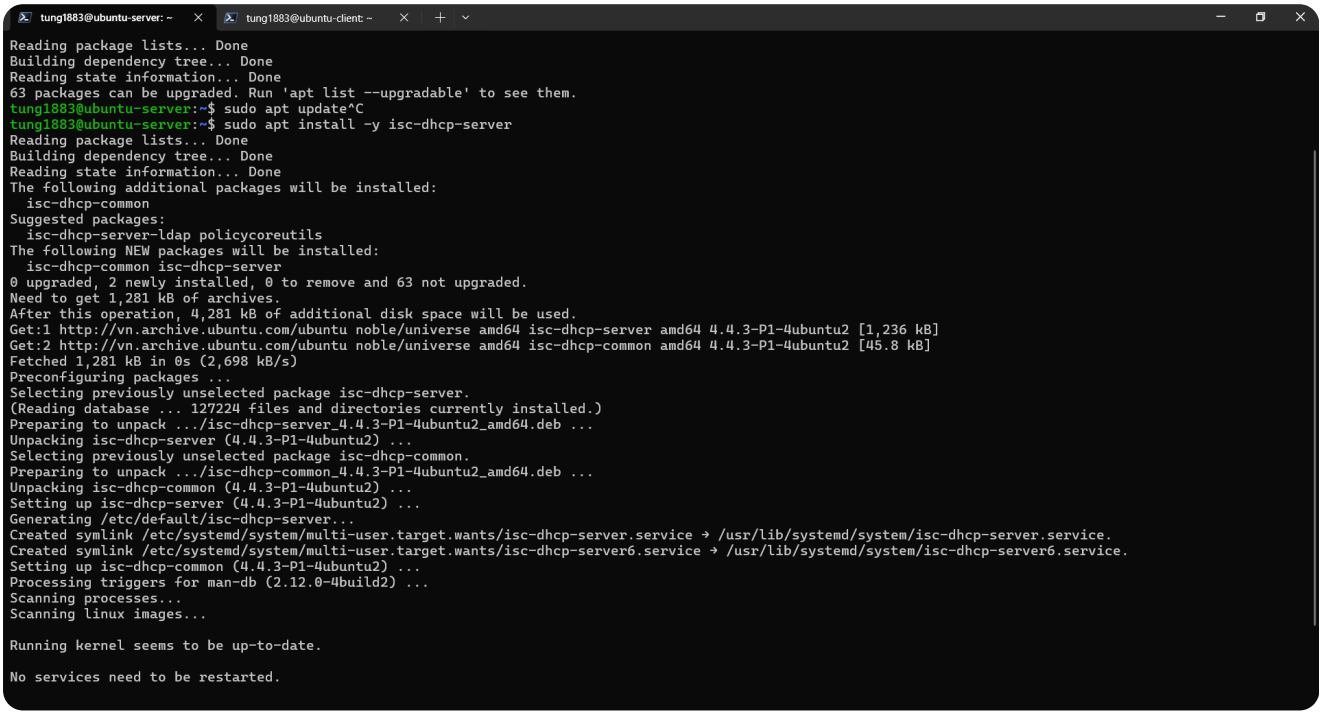


3. Cấu hình DHCP (trên server)

3.1 Cài đặt isc-dhcp-server

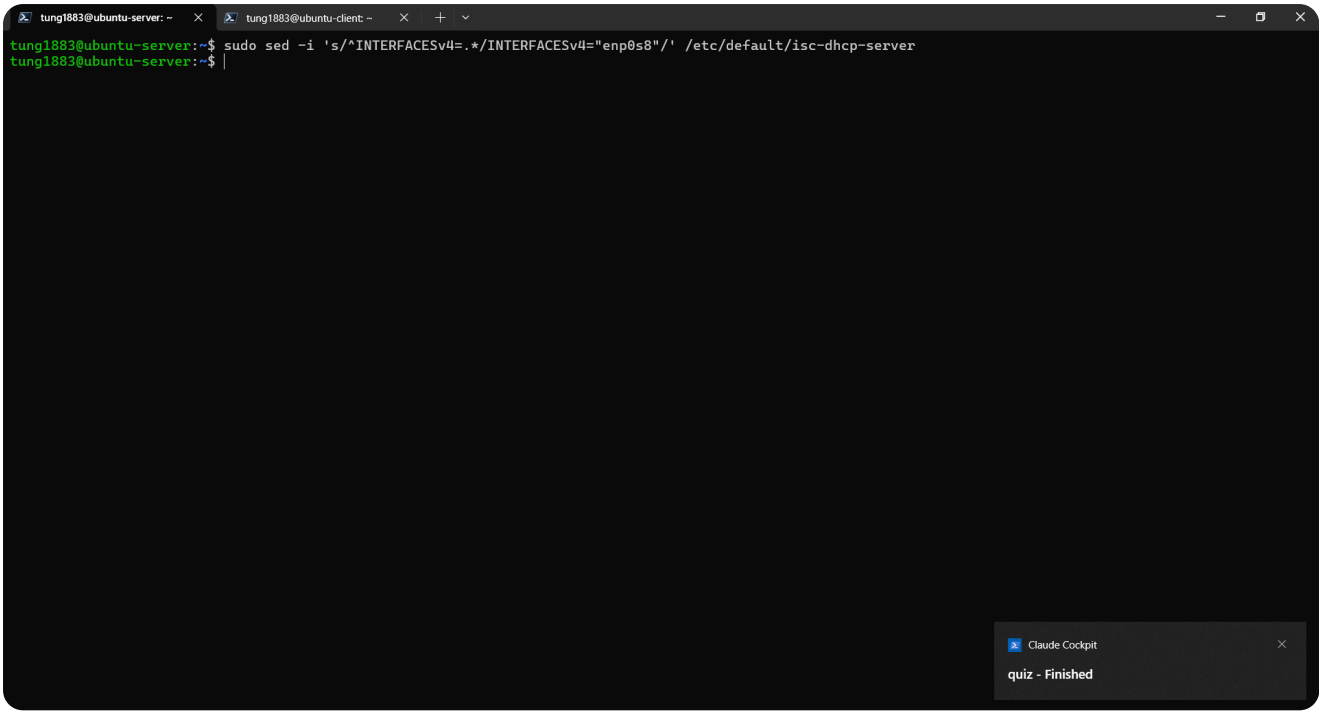
```
sudo apt update

sudo apt install -y isc-dhcp-server
```



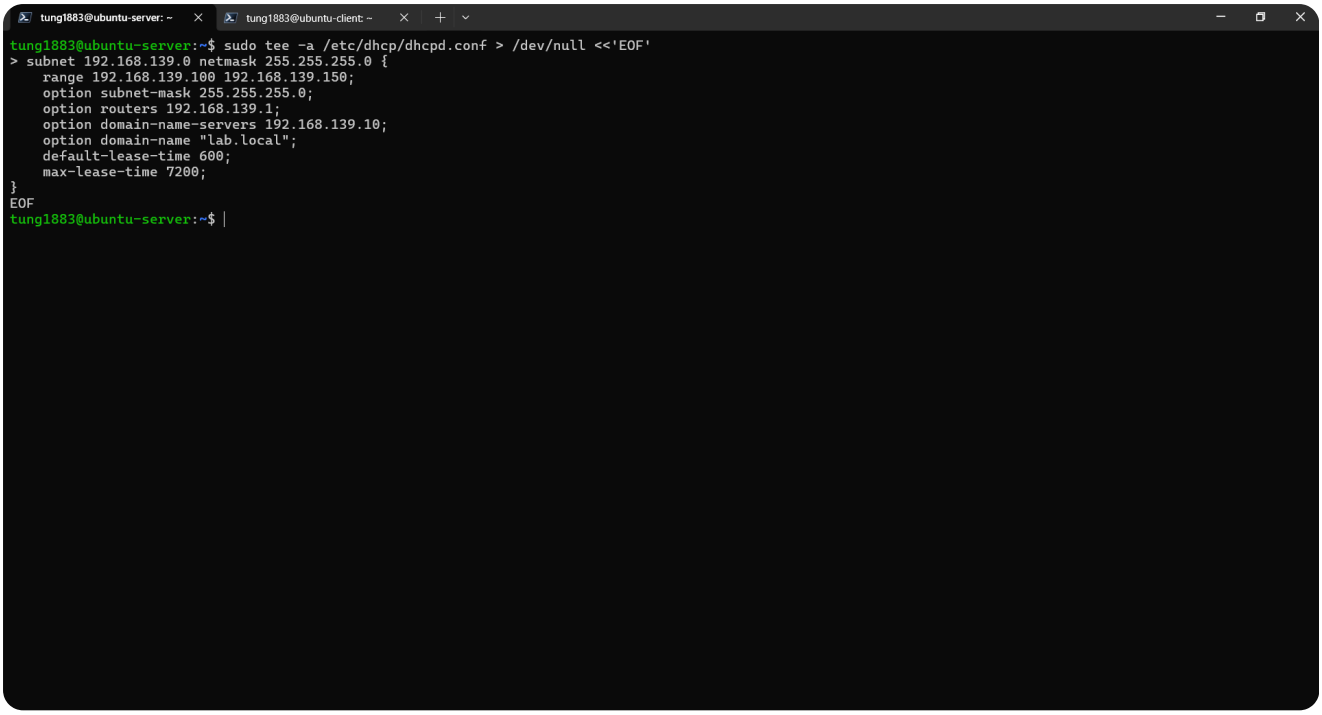
3.2 Chỉ định interface lắng nghe DHCP

```
sudo sed -i 's/^INTERFACESv4=.*/INTERFACESv4="enp0s8"/' /etc/default/isc-dhcp-server
```



3.3 Khai báo subnet, dải cấp phát, gateway, DNS, domain

```
sudo tee -a /etc/dhcp/dhcpd.conf > /dev/null <<'EOF'
subnet 192.168.139.0 netmask 255.255.255.0 {
    range 192.168.139.100 192.168.139.150;
    option subnet-mask 255.255.255.0;
    option routers 192.168.139.1;
    option domain-name-servers 192.168.139.10;
    option domain-name "lab.local";
    default-lease-time 600;
    max-lease-time 7200;
}
EOF
```

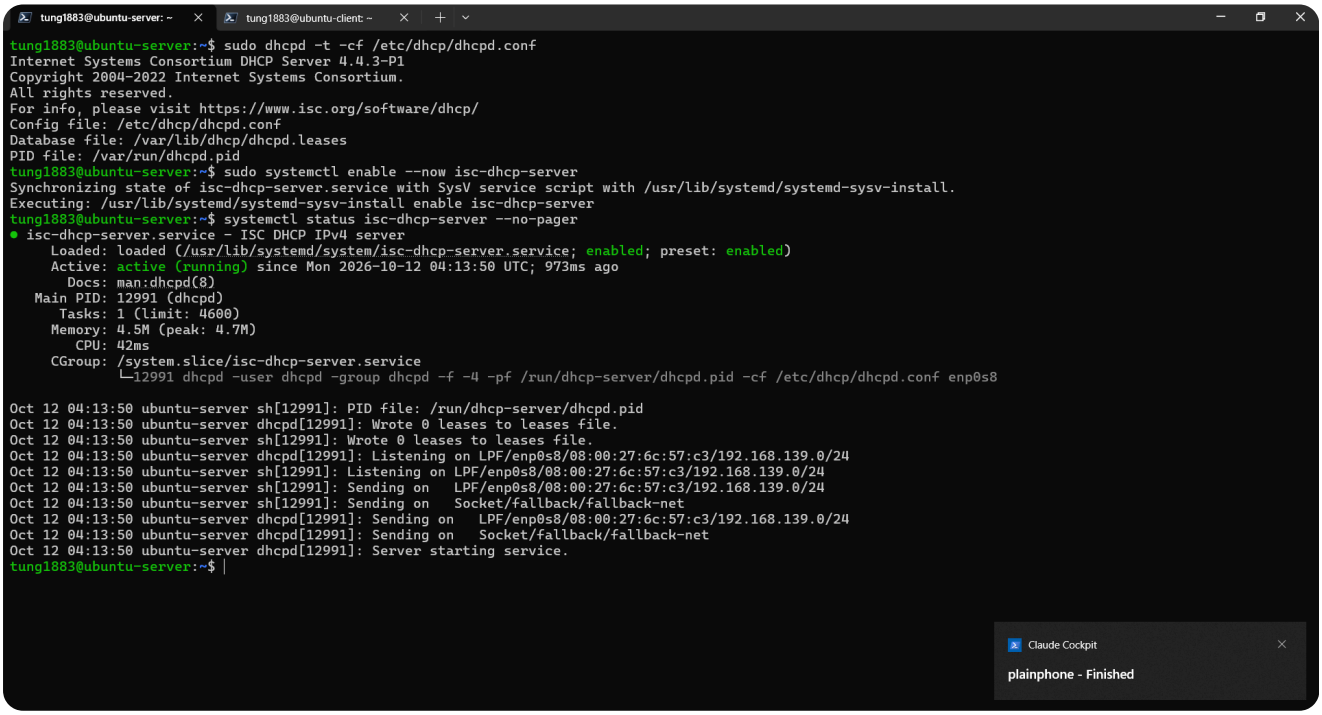


3.4 Kiểm tra cú pháp và khởi động dịch vụ

```
sudo dhcpd -t -cf /etc/dhcp/dhcpd.conf

sudo systemctl enable --now isc-dhcp-server

systemctl status isc-dhcp-server --no-pager
```

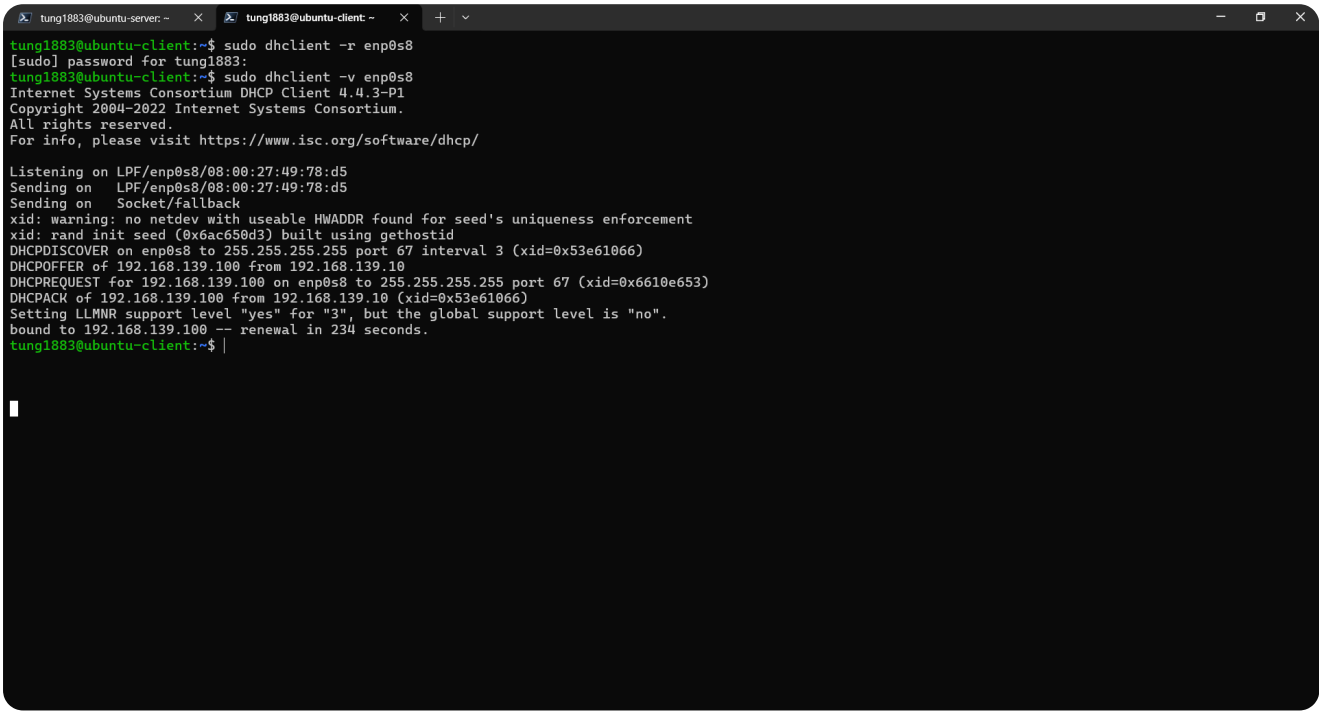


4. Kiểm tra DHCP (trên máy khách)

4.1 Giải phóng và yêu cầu cấp lại địa chỉ IP

```
sudo dhclient -r enp0s8

sudo dhclient -v enp0s8
```

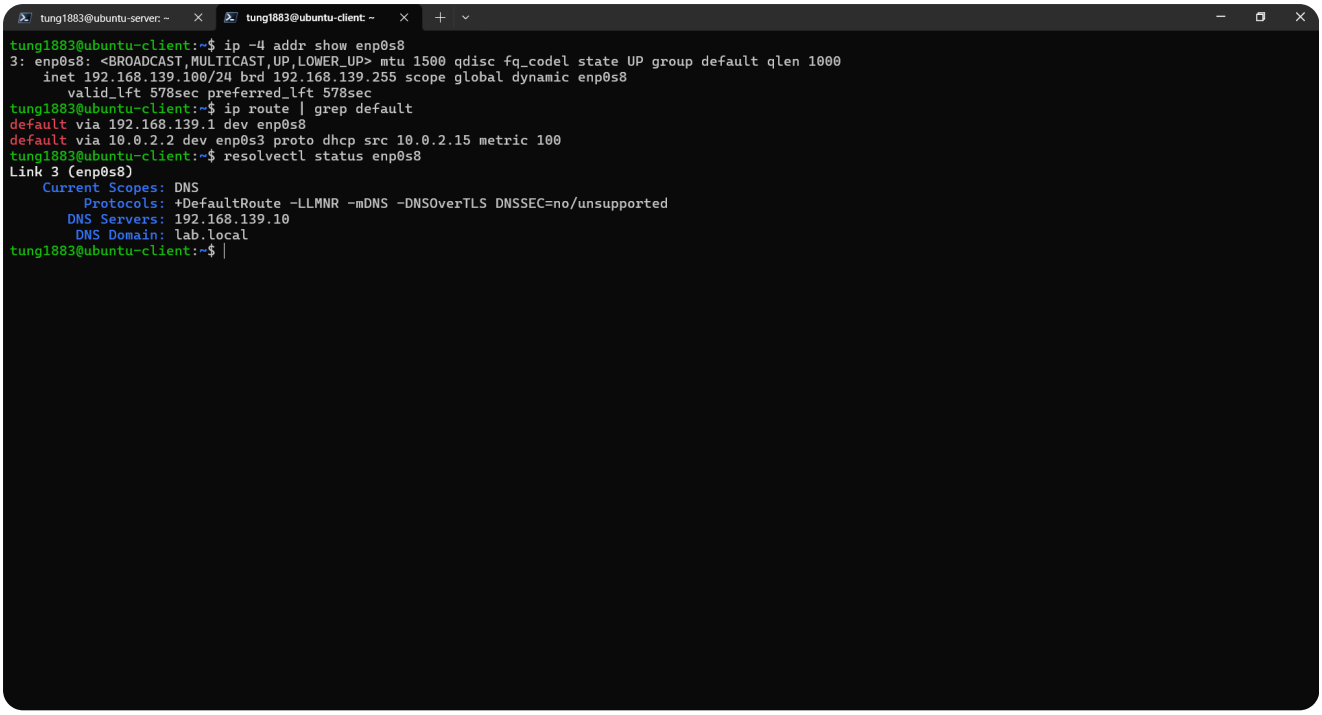


4.2 Kiểm tra IP, gateway, DNS nhận được

```
ip -4 addr show enp0s8

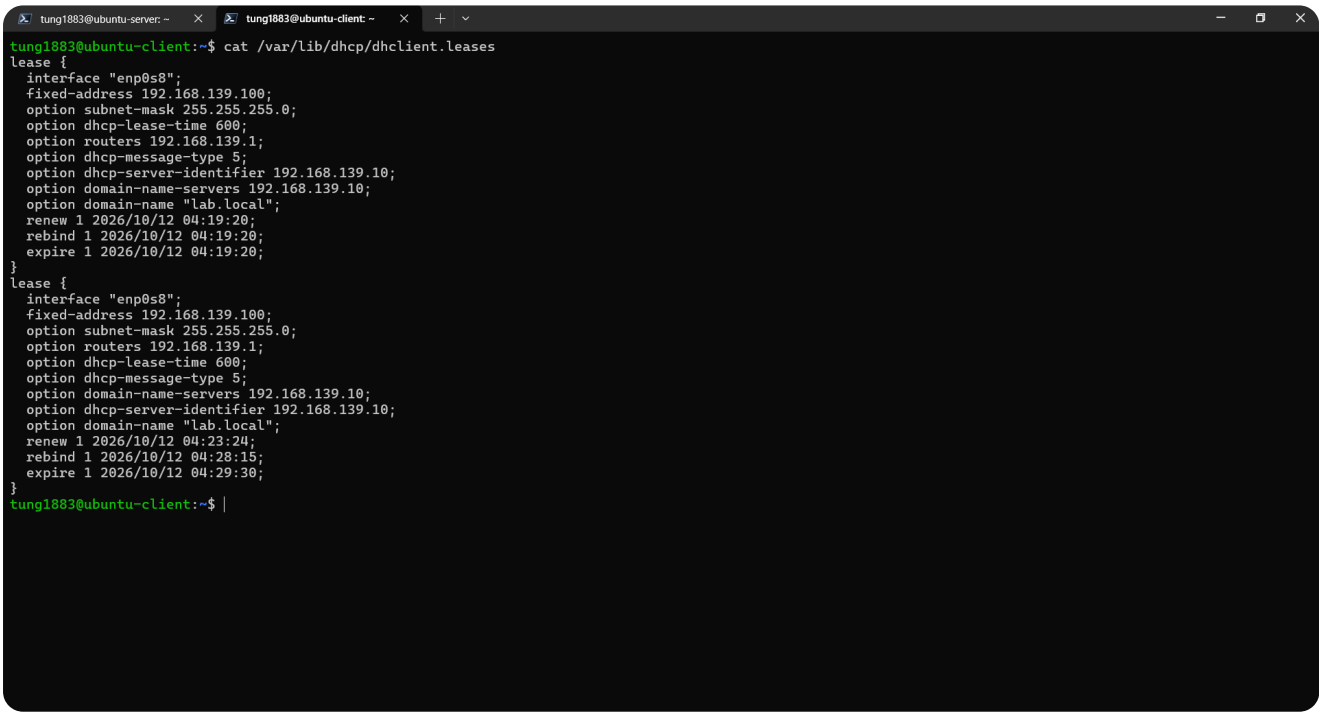
ip route | grep default

resolvectl status enp0s8
```



4.3 Xác định thời gian thuê địa chỉ

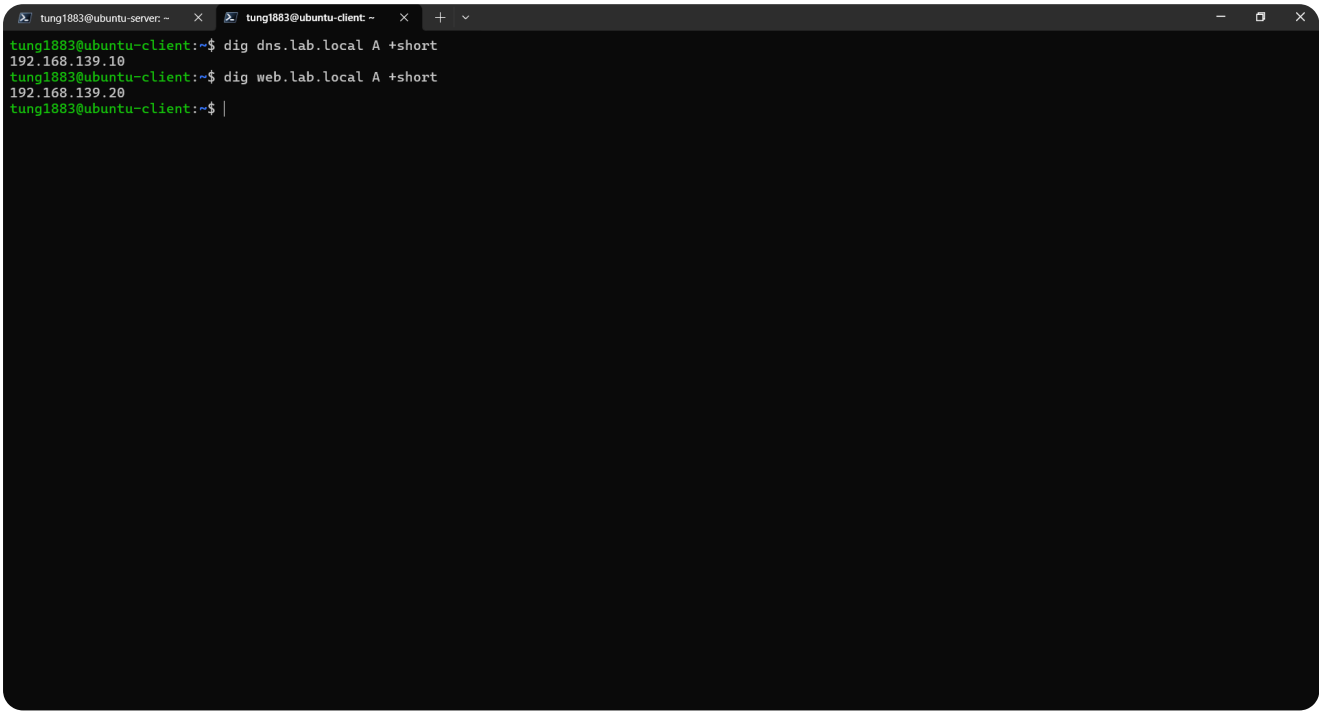
```
cat /var/lib/dhcp/dhclient.leases
```



4.4 Truy vấn các bản ghi DNS đã tạo (qua DNS server nhận từ DHCP)

```
dig dns.lab.local A +short

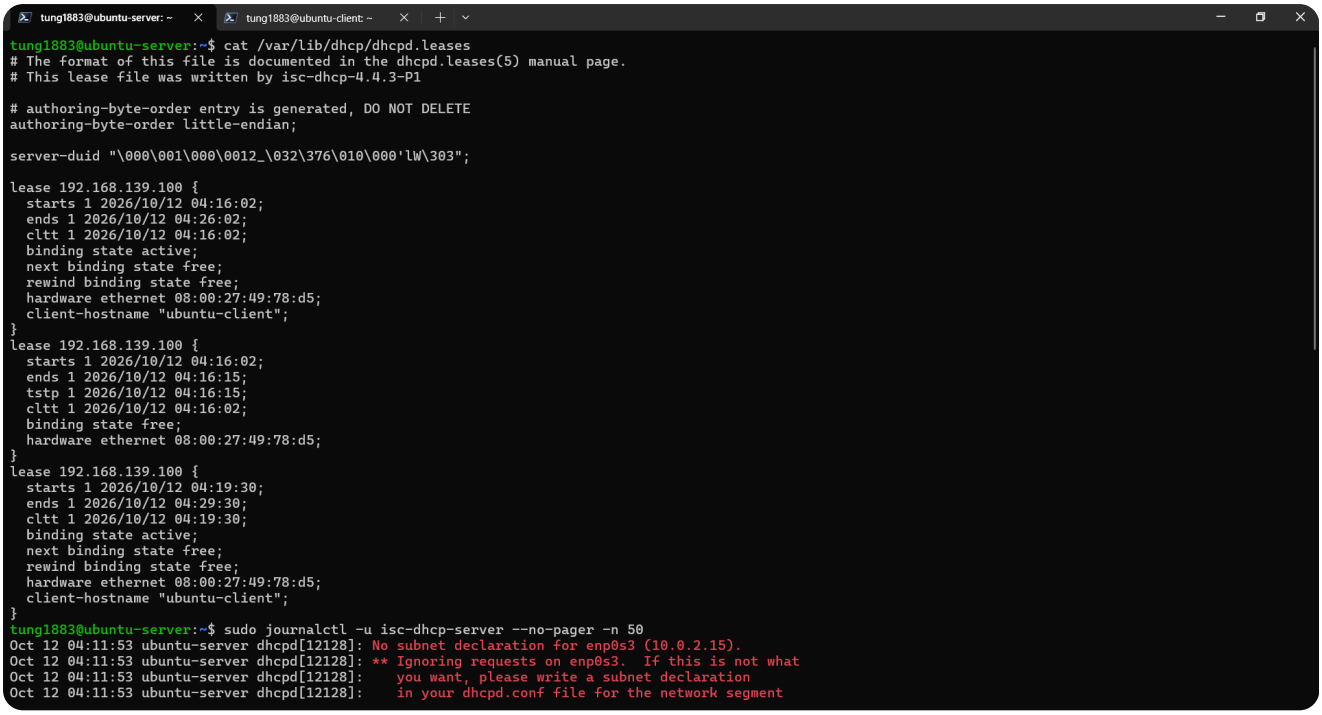
dig web.lab.local A +short
```



4.5 Kiểm tra thông tin cấp phát và nhật ký trên máy chủ DHCP

```
cat /var/lib/dhcp/dhcpd.leases

sudo journalctl -u isc-dhcp-server --no-pager -n 50
```


tung1883@ubuntu-server: ~
tung1883@ubuntu-client: ~
tung1883@ubuntu-server:~$ cat /var/lib/dhcp/dhcpd.leases
The format of this file is documented in the dhcpd.leases(5) manual page.
This lease file was written by isc-dhcp-4.4.3-P1
authoring-byte-order entry is generated, DO NOT DELETE
authoring-byte-order little-endian;
server-duid "\000\001\000\0012_\032\376\010\000'lW\303";
lease 192.168.139.100 {
starts 1 2026/10/12 04:16:02;
ends 1 2026/10/12 04:26:02;
cltt 1 2026/10/12 04:16:02;
binding state active;
next binding state free;
rewind binding state free;
hardware ethernet 08:00:27:49:78:d5;
client-hostname "ubuntu-client";
}
lease 192.168.139.100 {
starts 1 2026/10/12 04:16:02;
ends 1 2026/10/12 04:16:15;
tstp 1 2026/10/12 04:16:15;
cltt 1 2026/10/12 04:16:02;
binding state free;
hardware ethernet 08:00:27:49:78:d5;
}
lease 192.168.139.100 {
starts 1 2026/10/12 04:19:30;
ends 1 2026/10/12 04:29:30;
cltt 1 2026/10/12 04:19:30;
binding state active;
next binding state free;
rewind binding state free;
hardware ethernet 08:00:27:49:78:d5;
client-hostname "ubuntu-client";
}
tung1883@ubuntu-server:~$ sudo journalctl -u isc-dhcp-server --no-pager -n 50
Oct 12 04:11:53 ubuntu-server dhcpd[12128]: No subnet declaration for enp0s3 (10.0.2.15).
Oct 12 04:11:53 ubuntu-server dhcpd[12128]: ** Ignoring requests on enp0s3. If this is not what
Oct 12 04:11:53 ubuntu-server dhcpd[12128]: you want, please write a subnet declaration
Oct 12 04:11:53 ubuntu-server dhcpd[12128]: in your dhcpd.conf file for the network segment